is also defined as an extension to the untethered office concept, allowing unfettered access to the Internet or corporate networks any time, from any location.

An IP address uniquely identifies physical attachment to the subnet of Internet connectivity. When a source and a destination communicate over the Internet, they generally employ TCP for an end-to-end logical connection. When this logical connection is created, a TCP port number is assigned at both source and destination ends so that these two hosts remain in connection. Every end-to-end TCP logical connection is uniquely identified by four values: source IP address, source TCP port, destination IP address and destination TCP port. These remain fixed and static for the duration of the connection.

The basic problem with mobile node (MN) is that it may move from network to network. Therefore the challenge in mobile IP is to manage MN's IP address while dynamically changing location so that MN can always communicate with its permanent IP address. The solution is made with the proposed working principle of mobile IP, illustrated in Figure 5.

A node that roams and changes its position dynamically but keeps on communicating with any other system in the Internet as long as link-layer connectivity is given is defined as MN. Home network (HN) is the subnet MN longs for. MN has its IP address with the home subnet. No mobile IP support is needed when MN communicates within HN. Foreign network (FN) is the subnet except HN. The current subnet is that in which MN node is visiting currently.

MNs are assigned two IP addresses. The first is called home address. Home IP address is a static address used to identify the end-to-end connection, and is used by MN when connected to its HN.

The second address is care of address (COA). It is not static. It is a dynamic address used for packet routing only and is used by MN by FN. CAO changes every time MN changes to a new sub-network. All IP datagrams sent to MN are delivered to COA, and not directly to the static IP address when MN is in FN.

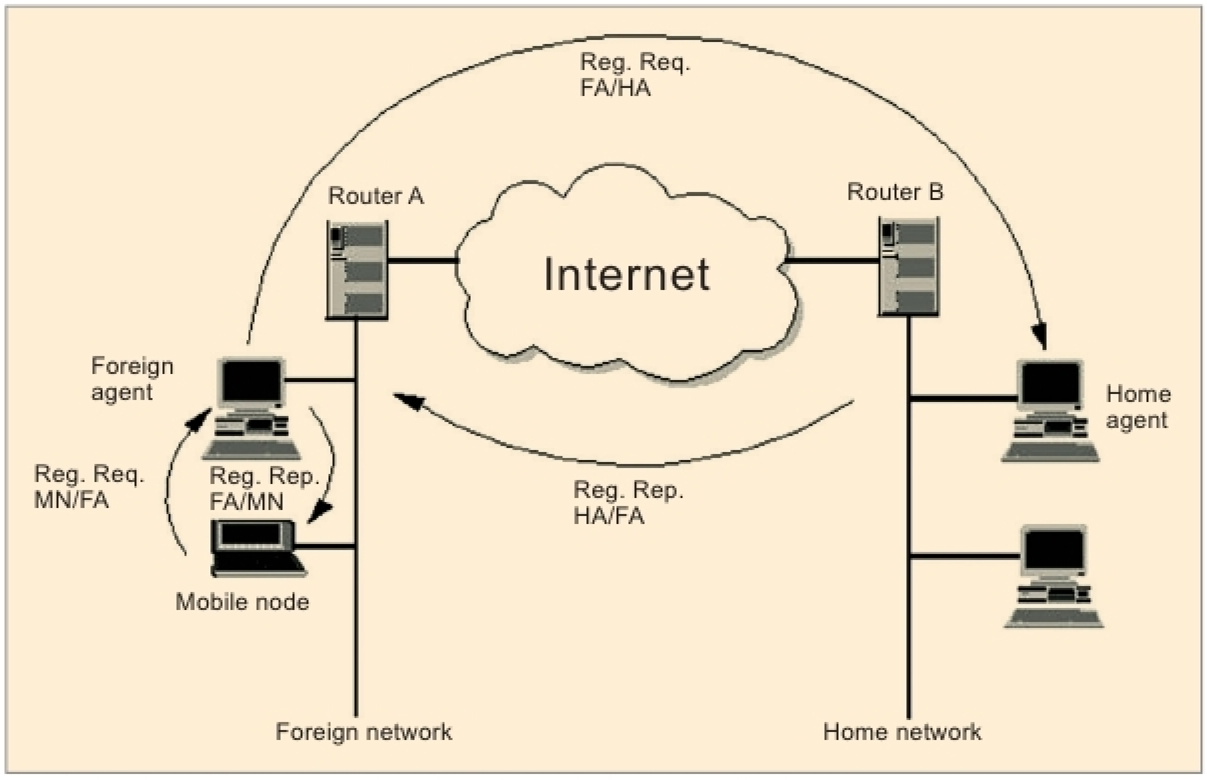

Figure 5: Illustration of connection while mobile node (MN) is in roaming state

There are two agents: foreign agent and home agent. These are known as mobility agents. These are mobile-IP-aware servers or routers that know where MN is actually connected. Home agent is HN's mobile IP agent, which has the responsibility of forwarding MN's packets to FN where MN is actually connected.

Foreign agent is responsible for delivering packets to the transient MN. It provides many services to MN during its visit to FN. It tracks COA, acting as the tunnel endpoint and forwarding packets to MN. Foreign agent is located in FN. It is the default router for MN when it is in FN. It also provides security services to MN.

Home agent is located in HN. It forwards IP packets to MN via COA. It maintains linked coordination between the home IP address of MN and its COA. A tunnel is established by home agent between itself and a reachable point (usually COA) for MN when it is in FN.

Working principle of mobile IP: The working principle of routing IP datagram in fixed Internet makes use of routers. Routers make use of IP address of IP datagram to route data from a source node to destination node.

IP address has two parts: network ID and host ID. Network ID identifies the sub-network on which the host is connected. Host ID is the host identification number.

Network ID is used by routers to move the datagram from one sub-network to another. Final router in the target sub-network moves the packet to a particular host based on host ID. MN communicates with another host or node on the Internet, mobile or not. The entire mobile IP process is transparent. MN is known to any other node or host by home address of MN wherever MN is physically present.

All packets between any other node or host and MN use MN's home address, regardless of whether MN is in HN or FN. COA is only used for communication with mobility agents and is never seen by any node or host while communicating with MN.

One of the important design goals of mobile IP is to make mobility possible without major changes to the large, existing installed base of IPv4,